

Characteristics		3G		WIFI	
Receive Message	Message / Hour	1,708	160,278	3,628	263,314
	Percent Battery / Hour	18.43%	16.13%	3.45%	4.23%
	Percent Battery / Message	0.01709	0.00010	0.00095	0.00002
	Messages Received (Note the losses)	240/1024	1024 / 1024	524 / 1024	1024 / 1024
Send Message	Messages / Hour	1,926	21,685	5,229	23,184
	Percent Battery / Hour	18.79%	17.80%	5.44%	3.66%
	Percent Battery / Message	0.00975	0.00082	0.00104	0.00016

Figure 3: MQTT vs HTTP

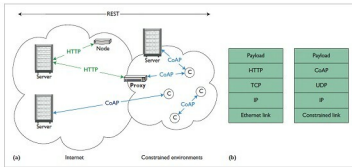


Figure 4: The CoAP model

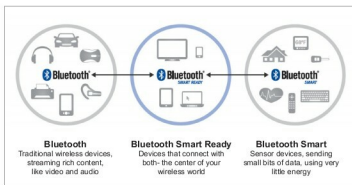


Figure 5: Bluetooth flavours

the wider Internet using similar protocols. CoAP is designed for use between devices on the same constrained network, between devices and general nodes on the Internet, and between devices on different constrained networks joined by the Internet. It is an application layer protocol designed for network constrained IoT devices like wireless sensor network nodes, and is often termed the lightweight version of HTTP with support for REST APIs. It can run on most devices that support UDP or a UDP analogue. It implements the REST architectural style which can be transparently mapped to HTTP. However, CoAP also provides features that go beyond HTTP such as native push notifications and group communication. While a usual HTTP header can be around 100 bytes, a CoAP standard header can be as light as just 4 bytes. Unlike MQTT, CoAP doesn't require a broker server to function. On the implementation side, the Eclipse Californium project provides a Java implementation of the CoAP protocol, including support for the DTLS security layer. There's also a MicroCoAP project which provides CoAP implementation for Arduino. Check out <https://github.com/1248/microc>.

3. Bluetooth and Bluetooth Low Energy

While MQTT and CoAP are infrastructure-independent, which means that it doesn't matter whether you're connected to a wired or a wireless network, Bluetooth provides only wireless communication over radio frequency (2.4GHz spectrum in the ISM band) using an industry standard that was initially used to share files between mobile phones and is now powerful enough to play music (Advanced Audio Distribution Profile/A2DP), stream data, or build your next IoT device. Bluetooth, generally, is divided into three categories.

Bluetooth Classic: This is meant for high data rate applications like streaming audio wirelessly.

Bluetooth Smart or Low Energy/BLE: This is meant for low powered battery-operated devices that stream low packets of data.

Bluetooth SmartReady: These are essentially the 'hub' devices such as computers, smartphones, etc. They support both the 'classic' and 'smart' devices.

Bluetooth is a sophisticated *ad hoc* networking protocol, and is now especially designed from the ground up for IoT. It provides a stable connection and communication channel, which is extremely low profile and low powered. An obvious example is fitness trackers, which even though powered on throughout the day, can last for months on a single charge or run on a coin cell battery, all thanks to BLE (Bluetooth Low Energy). Bluetooth Classic has fixed profiles like UART over Bluetooth class and A2DP class for audio streaming. On the other hand, Bluetooth Low Energy provides GATT or Generic Attribute Profile, which allows users to define their own profile using Bluetooth, like in the case of a heart rate monitor. BLE is extremely flexible and useful in the IoT space. Bluetooth 5.0 is already out and is maturing, offering more range, more data rates and double the transmission speeds.

Which protocol should I use for my next IoT project?

There are many different protocols and industry standards that are specially designed for IoT and can be used for it, such as the few mentioned above and others like Wi-Fi WebSockets, Zigbee, LoRA, Simple RF, XMPP, RFID, NFC, etc. Yet, one's choice should be based on the project requirements and the constraints of the application you are thinking of developing. MQTT, for example, is extremely powerful when you have an actuator network which needs to respond to a common sensor. The PUB/SUB model is ideal in that case. In the case of CoAP, you can create your own constrained network environment and relay information to the Internet via proxy. If your project does not require the Internet or long range communication, like a fitness tracker, then Bluetooth Low Energy could be the best choice. The possibilities in the IoT space are endless.

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